Derivative Securities FNCE 30007

Tutorial 2

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Review

Futures contract can be used for arbitrage, speculate, and hedge. Hedger's goal is taking a futures position to minimize the risk.

Long and Short Hedge:

- Long hedge:
  - Want to buy assets some time in the future and lock in the price.
- Short hedge:
  - Want to sell assets some time in the future and lock in the price.
  - It is appropriate when the hedger already owns an asset (or will own it at some definite date) and expects to sell it at some time in the future.

If the hedger gain (loss) in the spot market, he/she will lose (gain) in the futures market.

- Minimize the risk.

Review

Short Hedge.

A short hedge is used by producers (e.g. corn farmers) who are long cash inventory and fear price declines. To lock in a known price:

Example Setup

- Farmer expects to harvest 50,000 bushels in June.
- On April 20, spot = \$3.50/bu; June futures = \$3.35/bu.
- Farmer shorts 10 futures contracts (5,000 bu per contract) at \$3.35/bu to lock in that price.

Review

Scenario 1: Spot falls to \$3.10/bu

- Spot cash sale: $\$3.10 \times 50{,}000 = \$155{,}000$
- Futures gain: $(\$3.35 - \$3.10) \times 50{,}000 = \$12{,}500$
- Total received $= \$167{,}500 \Rightarrow$ Effective price $= \$3.35$/bu

Scenario 2: The spot rises to \$3.70/bu

- Spot cash sale: $\$3.70 \times 50{,}000 = \$185{,}000$
- Futures loss: $(\$3.70 - \$3.35) \times 50{,}000 = -\$17{,}500$
- Total received $= \$167{,}500 \Rightarrow$ Effective price $= \$3.35$/bu

Regardless of spot price movement, the hedged revenue per bushel matches the locked-in futures price. The farmer has effectively eliminated price uncertainty. Basis risk and transaction costs aside, the strategy achieves price certainty.

Review

Basis: Spot price of asset to be hedged − Futures price of contract used. (this should be valued at the same time)

- Before expiration, the basis may be positive or negative.
- When the spot price increases by more than the futures price, the basis increases. We call this strengthening of the basis.
- When the futures price increases by more than the spot price, the basis decreases. We call this weakening of the basis.

Review

Suppose one would like to buy (sell) an asset at time 1, he enters into futures market by taking a long (short) position with futures price F_0

- Suppose that he close out the future position exactly at time 1.
- $Cost(Revenue) = \underbrace{S_1}_{\text{cost(rev) for buy (sell) asset}} + \underbrace{F_0 - F_1}_{\text{cost (rev) from closing out}}$ for long (short).
- $Cost(Revenue) = F_0 + \underbrace{S_1 - F_1}_{\text{basis at time 1}}$

Cross hedge with different asset: Then we will have F_0^*, F_1^*, * refers to the asset is different from the asset that we want to hedge.

Review

Hedge ratio: The ratio of the size of the hedge contract to the size of the exposure. Given hedge ratio h

- $Cost(Revenue) = S_T + h * F_0^* - h * F_T^*$
- As the change in the value per unit of the asset ΔX, the change in value of the hedged position is:
  $\Delta Cost(Revenue) = \Delta S_T + h * F_0^* - h * \Delta F_T^*$
- Find h to minimize the variance of previous formula.
- $h^{min} = \frac{cov(\Delta S_T, \Delta F_T^*)}{var(\Delta F_T^*)} = \rho \frac{\sigma_S}{\sigma_{F^*}}$
  - h^{min}: minimum variance hedge ratio
- Q_A: size of the position being hedged in units
- Q_F: size of one futures contract in units.
- $N_{min} = \frac{h^{min} * Q_A}{Q_F}$: minimum variance number of the futures contract.

Review

Hedging with Stock Index Futures This is often used when you have a large equity portfolio and wish to reduce the market exposure of this position.

- V_A: The dollar value of the portfolio
- V_F: The dollar value of one futures contract
- Suppose you want a total beta $\tilde{\beta}$ for your portfolio with current β.
  - $(\beta - \tilde{\beta}) * \frac{V_A}{V_F}$

Tutorial Question

Q1: Does a perfect hedge always succeed in locking in the current spot price of an asset for a future transaction? Explain your answer.

- No.
- When you enter into a futures contract to hedge, you lock in the current futures price for a future transaction, not the current spot price.
- Suppose that the spot price of oil today is \$100 per barrel and the futures price for delivery in six months is \$105 per barrel. You enter into a futures contract to hedge against price fluctuations.
  - Outcome: You lock in the futures price of \$105, not the spot price of \$100.
  - At the end of six months, you will effectively pay \$105 per barrel regardless of the spot price at that time.

Tutorial Question

Q2: Explain why a short hedger's position improves when the basis strengthens unexpectedly and worsens when the basis weakens unexpectedly.

- A short hedge is long the asset and short the futures contract. The basis is the difference between the spot price and the futures price:

$$B_t = S_t - F_t$$

- The basis strengthens when the spot price increase more than or decreases less than the futures price. The value of the hedged position for the short hedger can be written as:

$$Value = F_0 + B_t$$

  which increases with the basis. Intuitively, the value of the asset has increased by more than the hedge contract, which benefits the short hedger.

Tutorial Question

Q3: Imagine you are the treasurer of a Japanese company exporting electronic equipment to the U.S. Discuss how you would design a foreign exchange hedging strategy and the arguments you would use to sell the strategy to your fellow executives.

Treasurer's task:

- Beaware that revenues are earned in U.S. dollars (USD), but costs and financial reporting are in Japanese yen (JPY).
- Estimate the cashflows in JPY and USD
- Enter into forward or futures contracts to lock in the foreign exchange rate for the USD cashflows

Tutorial Question

Q4: "For an asset where futures prices are usually less than spot prices, long hedges are likely to be particularly attractive." Explain this statement.

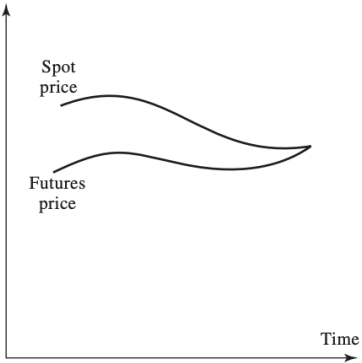


Figure: Backwardation

- Backwardation (Contango): A market condition where futures prices are lower (higher) than spot prices.
- Benefit: If spot prices remain the same or rise or even drop slightly, you gain by paying less for the asset than its spot value.

Tutorial Question

Suppose a seasonal food producer expects to sell gourmet olive oil in six months. Currently:

$$\text{Spot price } S_0 = \$200/\text{barrel}, \quad \text{Futures price } F_0 = \$190/\text{barrel}.$$

If the spot price S_T behaves like a random walk, then $\mathbb{E}[S_T] = S_0$. Under this assumption, suppose two outcomes are equally likely at expiration:

Scenario A: Supply Shortage Spot rises to \$220 → futures settles at \$220. Profit (long futures) = $220 - 190 = +\$30$.

Scenario B: Oversupply Spot falls to \$180 → futures settles at \$180. Loss (long futures) = $180 - 190 = -\$10$.

Under equally likely scenarios:

$$\mathbb{E}[\text{profit}] = \frac{(+30) + (-10)}{2} = +\$10/\text{barrel}.$$

Thus, in backwardation with a martingale spot, a long futures position yields positive average gains—even though volatility remains.

Tutorial Question

Q5: A corn farmer argues: "I don't use futures contracts for hedging. My real risk is not the price of corn. It is my whole crop gets wiped out by the weather." Discuss this viewpoint. Should the farmer estimate his or her expected production of corn and hedge to try to lock in a price for expected production?

- If weather creates a significant uncertainty about the volume of corn that will be harvested, the farmer should not enter into short forward contracts to hedge the price risk on his or her expected production. If the farmer cannot accurately predict their production, the farmer may find themselves under- or over-hedged depending on the weather situation.

Tutorial Question

Q5: Cont'd

- In this particular case, if the weather is bad and the farmer's production is lower than expected, they are over-hedged, implying they have a net short position in the futures market. However, other farmers are likely to have been affected similarly. Corn production overall will be low and as a consequence the price of corn will be relatively high. The farmer's problems arising from the bad harvest will be made worse by losses on the short futures position.
- In the opposite case of a very good harvest, the farmer will be under-hedged and prices will be lower than expected. The short futures contract at the relatively higher price provides some protection against this scenario, however, with better production, the farmer is likely to be in a better financial position than in the case of bad weather, and the former scenario should be the main factor (assuming they are risk-averse).

Tutorial Question

Q6: On July 1, an investor holds 50,000 shares of a certain stock. The market price is \$30 per share. The investor is interested in hedging against movements in the market over the next month and decides to use the September Mini S&P 500 futures contract. The index is currently 1,500 and one contract is for delivery of \$50 times the index. The beta of the stock is 1.3 (where beta is computed against the SP 500 index value). What strategy should the investor follow to achieve the minimum variance hedge ratio?

- The investor should take a short position.
- $N^{min} = \beta * \frac{50K*30}{1.5K*50} = 26$ contracts

Tutorial Question

Q7: After reading the Hershey case presented in the lecture, think about and answer the following two questions.

- In 2024, Hershey reported substantial gains from commodity derivatives while cocoa prices were rising sharply. Does this mean that Hershey's hedging program was designed to generate profits? Explain your answer.
- Although cocoa prices had fallen substantially by 2026, why might Hershey still face significant cocoa-related costs or report losses on some commodity derivative positions?

Hedging is symmetric — the same mechanism that shielded Hershey in 2024 necessarily prevents it from capturing the 2026 decline. Ex post regret in one direction is the unavoidable counterpart of ex post relief in the other.